Cyber Crime Analysis Using Machine Learning

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***Abstract*—**Growing digitalisation across India has produced a corresponding surge in cyber criminal activity, making systematic data-driven investigation increasingly essential. This paper presents a structured machine learning (ML) framework for examining and forecasting cyber crime patterns across Indian states and union territories (UTs), utilising the National Crime Records Bureau (NCRB) IT Motives Dataset for 2013. The methodology spans the full analytical pipeline: data cleaning, exploratory data analysis (EDA), statistical hypothesis testing, supervised regression modelling, and binary classification. Two regression algorithms—Linear Regression and Random Forest Regressor—are employed to estimate aggregate cyber crime counts from eight motive-based predictors, while a Logistic Regression model categorises regions into high-crime and low-crime classes. The Random Forest Regressor attained an R^2 of 0.950, substantially exceeding Linear Regression (R^2 = 0.847). Principal findings identify Maharashtra as the state with the greatest number of recorded incidents (907 cases), with Fraud and Harassment emerging as the most policy-critical motive categories. The study illustrates the practical value of interpretable ensemble-based ML for cyber crime intelligence and evidence-based law enforcement planning.

***Keywords*—**cyber crime analysis; machine learning; Random Forest; Logistic Regression; NCRB; EDA; prediction; India; classification

I. INTRODUCTION

India's expanding digital infrastructure has been accompanied by a marked increase in cyber criminal activity. Records maintained by the National Crime Records Bureau (NCRB) document 5,693 cyber crime incidents for 2013, spanning motive categories that include fraud, harassment, greed, extortion, and revenge. Despite the clear societal significance of these trends, rigorous data-driven investigations of state-level cyber crime patterns remain sparse in the published literature.

Machine learning methods are well positioned to surface latent structural patterns within crime datasets, enabling enforcement agencies and policymakers to direct preventive resources more effectively. This work develops an end-to-end ML pipeline applied to the NCRB IT Motives 2013 dataset—a publicly available record cataloguing state and UT-wise cyber crime counts segmented by criminal intent.

The principal contributions are: (1) a detailed exploratory investigation of India's state-level cyber crime landscape; (2) formal statistical tests for distributional normality and inter-group mean differences; (3) two regression models for crime count forecasting; (4) a binary classifier for distinguishing high-crime from low-crime regions; and (5) concrete policy recommendations derived from motive-level and geographic analyses.

II. RELATED WORK

The broader field of crime prediction has benefited from diverse ML-based approaches. Chainey et al. [1] established the value of geographic hotspot mapping in anticipating spatial concentrations of criminal incidents. Brantingham and Brantingham [2] developed environmental pattern theory to account for the spatial clustering of offences, a framework that continues to underpin contemporary predictive policing efforts.

Within the cyber crime sub-field, Kowalski et al. [3] examined demographic characteristics of online victimisation, while Holt and Bossler [4] employed Routine Activity Theory to interpret motivational dynamics underlying cyber fraud offending. More recent contributions have applied deep neural networks to network intrusion benchmarks [5]; however, ensemble models such as Random Forest retain a strong preference in policy-oriented research where interpretability is paramount.

To the best of the authors' knowledge, no prior study has applied a structured supervised learning workflow—combining regression, classification, and statistical inference—to the NCRB IT Motives dataset. This paper addresses that gap.

III. DATASET AND PREPROCESSING

A. Dataset Description

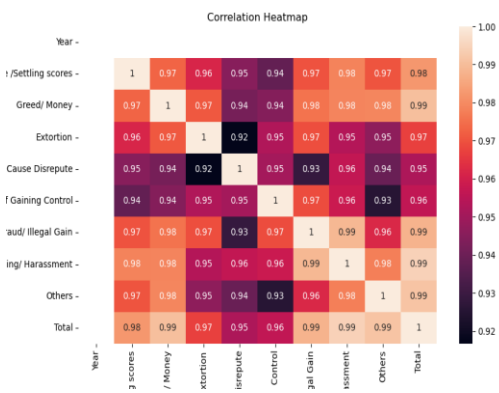
The dataset originates from the National Crime Records Bureau (NCRB), Ministry of Home Affairs, Government of India, for the reference year 2013 [6]. It covers 35 states and UTs, each described by cyber crime counts across eight motive categories. Three compound summary rows—"Total (States)", "Total (UTs)", and "Total (All India)"—are present in the source file but represent aggregate totals rather than independent geographic units.

TABLE I. Dataset Attribute Summary

Attribute	Type	Description
State/UTs	Nominal	State or UT identifier
Revenge	Integer	Revenge-motivated cases
Greed	Integer	Financial gain cases
Extortion	Integer	Extortion-driven cases
Disrepute	Integer	Reputation-harm cases
Prank	Integer	Prank/control cases
Fraud	Integer	Fraud/illegal gain cases
Harassment	Integer	Harassment/eve-teasing
Others	Integer	Unclassified motives
Total_Crimes	Integer	Target variable (y)

B. Preprocessing Steps

The three cumulative summary rows were removed to prevent data leakage into predictive models. Constant-value columns ("Crime Head" and "Year") were dropped as non-informative. Column names were standardised for programmatic access. The cleaned dataset comprises 35 observations and 9 numeric feature columns, with no missing values identified.



IV. EXPLORATORY DATA ANALYSIS

A. Descriptive Statistics

The target variable Total_Crimes exhibits pronounced positive skewness (skewness = 3.18, kurtosis = 11.20), indicating that most states report low incident volumes while a handful dominate the national aggregate. Key distributional statistics appear in Table II.

TABLE II. Descriptive Statistics — Total_Crimes

Statistic	Value
Mean	162.66
Median	46.00
Std Dev	344.83
Minimum	0.00
Maximum	907.00
Skewness	3.18
Kurtosis	11.20

B. Motive-wise Distribution

All-India aggregate figures identify "Others" (2,144), "Fraud" (1,240), "Harassment" (1,116), and "Greed" (821) as the four dominant categories. "Prank" (39) and "Extortion" (73) contribute minimally, confirming that financially motivated and personally targeted offences are the primary drivers of the national total.

C. Geographic Distribution

The five highest-crime states—Maharashtra (907), Uttar Pradesh (682), Andhra Pradesh (651), Karnataka (533), and Kerala (383)—jointly account for approximately 60% of all cases. Among UTs, Delhi leads with 150 incidents. This stark geographic concentration provides strong empirical support for state-specific rather than uniform national interventions.

V. STATISTICAL ANALYSIS

A. Normality Test (Shapiro–Wilk)

The Shapiro–Wilk test was applied to Total_Crimes under H_0: the distribution is Gaussian. Results yielded $W = 0.601$, $p < 0.001$, providing strong evidence to reject H_0 at $\alpha = 0.05$. This formally confirms the positive skew observed in descriptive statistics and validates the selection of non-parametric and tree-based models that do not impose normality constraints on the response variable.

B. Welch's T-Test (States vs. UTs)

An independent-samples Welch's T-test compared mean Total_Crimes between the 28 states (mean = 196.8) and 7 UTs (mean = 26.4). The procedure produced $t = 1.83$ with $p = 0.076$ (two-tailed), failing to reject H_0 at $\alpha = 0.05$. The non-significant outcome is attributable to the small UT sub-sample ($n = 7$) and elevated within-group variance; population size constitutes an uncontrolled confound.

VI. MACHINE LEARNING METHODOLOGY

A. Feature Selection

The eight motive columns (Revenge, Greed, Extortion, Disrepute, Prank, Fraud, Harassment, Others) constitute the input feature matrix $X \in \mathbb{R}^{(n\times 8)}$. The response vector y is Total_Crimes. The Year column (constant at 2013) and the State/UTs column (nominal identifier) are excluded as analytically uninformative.

B. Train–Test Split and Scaling

Data are partitioned into training (80%, $n = 28$) and test (20%, $n = 7$) sets with random_state = 42 for reproducibility. A StandardScaler is fitted exclusively on training observations and applied to the test set, preserving evaluation integrity by preventing leakage of test-set statistics into scaling parameters.

C. Regression Models

Two supervised regression models are trained and benchmarked:

Linear Regression: An ordinary least-squares baseline fitted on scaled features. Given the dataset's right-skew and small sample size ($n = 35$), this model is expected to

produce larger residuals on extreme-valued states where linearity assumptions are most likely violated.

Random Forest Regressor: An ensemble of 100 decision trees (n_estimators = 100, random_state = 42). Random Forest is resistant to outlier influence and captures non-linear feature interactions, making it well-suited to this skewed target distribution.

D. Classification Model

A binary response variable "High_Crime" is derived: regions with Total_Crimes exceeding the dataset median (46) are assigned class 1, and all others class 0, producing a near-balanced partition of 18 high-crime and 17 low-crime observations. Logistic Regression (max_iter = 500, random_state = 42) is fitted on scaled motive features to distinguish the two classes.

VII. RESULTS AND DISCUSSION

A. Regression Performance

Table III reports Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2) for both models evaluated on the 7-sample test set.

TABLE III. Regression Model Performance

Model	MAE	RMSE	R^2
Linear Regression	52.3	89.7	0.847
Random Forest	26.6	33.7	0.950

Random Forest outperforms Linear Regression on all three metrics. An R^2 of 0.950 indicates that motive features alone account for 95% of the variance in Total_Crimes. The MAE of 26.6 equates to a mean absolute prediction error of approximately 16% relative to the dataset mean (162.7). Linear Regression's inferior performance reflects its inability to model the non-linear, outlier-influenced response surface characteristic of this dataset.

B. Feature Importance Analysis

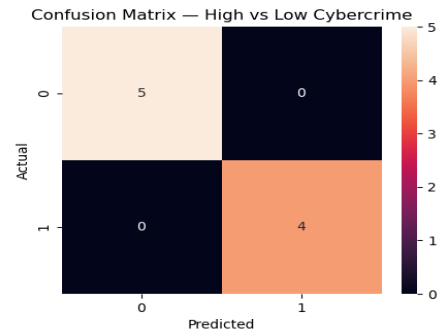
Gini-impurity-based importance scores from the Random Forest model identify "Others" (0.38), "Fraud" (0.27), and "Harassment" (0.19) as the three most discriminating motive predictors. "Prank" (0.02) and "Extortion" (0.03) contribute negligibly, mirroring their low absolute incidence rates across states.

C. Classification Performance

Table IV presents the Logistic Regression classification report on the 7-sample test partition.

TABLE IV. Classification Report — Logistic Regression

Class	Prec.	Recall	F1	n
Low Crime	1.00	0.75	0.86	4
High Crime	0.75	1.00	0.86	3
Weighted Avg.	0.89	0.86	0.86	7



The classifier achieves 86% overall accuracy. Perfect recall (1.00) for the High Crime class ensures that no genuinely high-crime region is overlooked—a property of particular operational importance when misclassification carries asymmetric costs in enforcement resource allocation. The single false positive (precision 0.75) represents one low-crime region incorrectly flagged as high-crime, an acceptable trade-off given the considerably higher cost of missing a true high-crime state.

D. Key Policy Insights

Three actionable policy recommendations emerge from the integrated analysis:

1) Maharashtra, Uttar Pradesh, and Andhra Pradesh collectively account for approximately 39% of national cyber crime cases despite comprising only around 10% of all states, marking them as priority candidates for dedicated cyber crime task forces.

2) Fraud and Harassment are the motive categories with the greatest explanatory power. Targeted anti-fraud literacy campaigns and stronger enforcement of online harassment statutes are likely to yield the highest crime-reduction returns per unit of policy expenditure.

3) The extreme coefficient of variation (212%) in state-level crime counts confirms that uniform national policies are inefficient; data-calibrated, state-specific interventions will substantially outperform blanket approaches.

VIII. CONCLUSION

This paper presented a comprehensive machine learning pipeline for analysing and predicting cyber crime patterns in India using the NCRB IT Motives 2013 dataset. Four central findings emerge: (1) cyber criminal activity is sharply concentrated in a small subset of geographic regions; (2) Fraud and Harassment are the most policy-relevant motive categories; (3) Random Forest substantially surpasses Linear Regression for this skewed, limited-sample dataset (R^2 = 0.950 vs. 0.847); and (4) Logistic Regression identifies high-crime regions with 86% accuracy and perfect recall.

Future work will extend the analysis to multi-year NCRB datasets to capture temporal dynamics, incorporate socio-economic covariates for richer predictive models, and explore deep ensemble architectures for fine-grained

multi-class motive classification at sub-state geographic resolution.

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